

Tier 6: Group-Based Trajectory Modelling (GBTM)

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0.1 Overview

This report implements a population-level GBTM following Yang et al. (2022, J Intensive Med). The earlier temporal report used a within-patient HMM that clusters states within one patient rather than trajectory shapes across patients. A latent-class growth model is fit on repeated 24-hour SOFA scores via flexmix, and class membership is then linked to mortality via Cox regression and to incident Sepsis-3 via logistic regression.

0.2 Load Libraries

```
suppressPackageStartupMessages({  
  library(flexmix); library(survival)  
})  
set.seed(486649)
```

0.3 Locate and Read Data

```
find_data <- function(f) {  
  cands <- c(file.path("../processed_data", f),  
             file.path("processed_data", f), f)  
  hit <- cands[file.exists(cands)][1]  
  if (is.na(hit)) stop("cannot find ", f); hit  
}  
OUT <- dirname(find_data("08_panel_sample_long.parquet"))  
FIG <- if (dir.exists("figure")) "figure" else { dir.create("figure"); "figure" }
```

```
panel <- as.data.frame(arrow::read_parquet(find_data("08_panel_sample_long.parquet")))
cohort <- as.data.frame(arrow::read_parquet(find_data("02_cohort.parquet")))[, c("stay_id", "age",
                                                                              "hospital_expire_flag", "los_hours")]
```

0.4 Build the First-24h SOFA Trajectory Sample

The panel is restricted to the first 24 hours, keeping only stays with at least 12 of the 25 possible hourly rows for a stable polynomial fit.

```
d <- panel[panel$hour <= 24, c("stay_id", "hour", "sofa_total")]
n_obs <- table(d$stay_id)
keep_ids <- names(n_obs[n_obs >= 12])
d <- d[as.character(d$stay_id) %in% keep_ids, ]
d$stay_id <- factor(d$stay_id)
cat(sprintf("GBTM sample: %d stays, %d stay-hours (first 24h, >=12 obs/stay)\n",
            length(unique(d$stay_id)), nrow(d)))
```

```
## GBTM sample: 2438 stays, 58267 stay-hours (first 24h, >=12 obs/stay)
```

0.5 Fit Latent-Class Growth Models

Latent-class growth models are fit for $k = 2$ through 4 classes, with the best selected by BIC. The `stay_id` syntax in `flexmix` assigns all rows of one stay to the same latent class (the GBTM formulation). Multiple random restarts at each k guard against local optima.

```
# `minprior` prunes near-empty components during EM; stepFlexmix
# discards restarts that degenerate and keeps clean convergences.
cat("fitting latent-class growth models (k=2..4; this takes a minute) ...\n")
```

```
## fitting latent-class growth models (k=2..4; this takes a minute) ...
```

```
sf <- stepFlexmix(sofa_total ~ poly(hour, 2) | stay_id, data = d, k = 2:4,
                  nrep = 3, control = list(iter.max = 200, minprior = 0.05),
                  verbose = FALSE)
```

```
## Error in FLXfit(model = model, concomitant = concomitant, control = control, :
## 136 Log-likelihood: NA
```

```
## Error in FLXfit(model = model, concomitant = concomitant, control = control, :
## 118 Log-likelihood: NA
```

```
cat("\nBIC by k:\n"); print(sf)
```

```
##
```

```
## BIC by k:
```

```
##
```

```
## Call:
```

```
## stepFlexmix(sofa_total ~ poly(hour, 2) | stay_id, data = d, control = list(iter.max = 200,
## minprior = 0.05), k = 2:4, nrep = 3, verbose = FALSE)
```

```
##
```

```
## iter converged k k0 logLik AIC BIC ICL
## 2 24 TRUE 2 2 -115455.2 230928.3 231009.1 231030.3
```

```
## 3    29      TRUE 3   3 -101903.3 203834.6 203960.2 203996.7
## 4    30      TRUE 3   4 -101903.3 203834.6 203960.3 203996.9
```

```
best <- getModel(sf, which = "BIC")
K <- ncol(best@posterior$scaled)
cat(sprintf("Selected K = %d classes by BIC\n", K))
```

```
## Selected K = 3 classes by BIC
```

```
bic_tab <- data.frame(k = 2:4, BIC = BIC(sf), selected = (2:4) == K)
write.csv(bic_tab, file.path(OUT, "18_gbtm_bic.csv"), row.names = FALSE)
```

0.6 Per-Stay Class Assignment

Each stay is assigned to the trajectory class with the highest posterior probability. Classes are reordered by mean SOFA severity so that class 1 always corresponds to the mildest trajectory.

```
cls_per_row <- clusters(best)
row_id <- d$stay_id
cls_by_stay <- tapply(cls_per_row, row_id, function(x) x[1]) # constant within stay
post <- best@posterior$scaled
post_max <- tapply(apply(post, 1, max), row_id, function(x) x[1])

assign_tab <- data.frame(stay_id = as.integer(names(cls_by_stay)),
                        class = as.integer(cls_by_stay),
                        posterior_prob = round(as.numeric(post_max), 4))

# Reorder by severity so class 1 = mildest (flexmix numbering is arbitrary).
sev_order <- order(tapply(d$sofa_total, cls_by_stay[as.character(d$stay_id)], mean))
remap <- setNames(seq_along(sev_order), sev_order)
assign_tab$class <- remap[as.character(assign_tab$class)]
write.csv(assign_tab, file.path(OUT, "18_tier6_gbtm_assignments.csv"), row.names = FALSE)
```

0.7 Mean Trajectory Per Class and Trajectory Plot

```
d$class <- remap[as.character(cls_by_stay[as.character(d$stay_id)])]
means <- aggregate(sofa_total ~ hour + class, data = d, FUN = mean)
means <- means[order(means$class, means$hour), ]
write.csv(means, file.path(OUT, "18_tier6_gbtm_class_means.csv"), row.names = FALSE)

png(file.path(FIG, "18_gbtm_trajectories.png"), width = 1100, height = 850, res = 200)
cols <- c("steelblue", "darkorange", "firebrick", "purple")[1:K]
plot(NULL, xlim = c(0, 24), ylim = range(means$sofa_total),
     xlab = "ICU hour", ylab = "Mean SOFA total",
     main = sprintf("GBTM: %d latent SOFA trajectory classes (first 24h)", K))
for (k in 1:K) {
  m <- means[means$class == k, ]
  lines(m$hour, m$sofa_total, col = cols[k], lwd = 2, type = "b", pch = 19)
}
legend("topleft", paste("Class", 1:K, c("(mildest)", rep("", K - 2), "(most severe)")[1:K]),
```

```
col = cols, lwd = 2, bty = "n", cex = 0.75)
dev.off()
```

```
## pdf
## 2
```

0.8 Link Trajectory Class to Mortality via Cox Regression

Following Yang et al. (2022), a Cox proportional hazards model links trajectory class membership to in-hospital mortality, with hospital length of stay (hours) as the time-to-event variable.

```
surv_d <- merge(assign_tab, cohort, by = "stay_id")
surv_d$class <- factor(surv_d$class, levels = 1:K)
surv_d$los_hours <- pmax(surv_d$los_hours, 0.5) # avoid zero survival time
cox_fit <- coxph(Surv(los_hours, hospital_expire_flag) ~ class + age, data = surv_d)
cox_s <- summary(cox_fit)
cox_tab <- data.frame(term = rownames(cox_s$coefficients),
                     HR = round(exp(cox_s$coefficients[, "coef"]), 3),
                     p_value = signif(cox_s$coefficients[, "Pr(>|z|)"], 3))
cat("\nCox model: trajectory class vs in-hospital mortality (ref = mildest class):\n")
```

```
##
## Cox model: trajectory class vs in-hospital mortality (ref = mildest class):
print(cox_tab, row.names = FALSE)
```

```
##      term      HR  p_value
## class2 2.525 2.22e-03
## class3 5.019 3.88e-08
##      age 1.029 1.74e-09
```

```
write.csv(cox_tab, file.path(OUT, "18_tier6_mortality_cox.csv"), row.names = FALSE)
```

0.9 Link Trajectory Class to Incident Sepsis-3

As a secondary analysis, logistic regression tests whether the SOFA trajectory shape over the first 24 hours associates with ever meeting Sepsis-3 criteria, beyond endpoint severity alone.

```
onset_d <- panel[!duplicated(panel$stay_id), c("stay_id", "sepsis3")]
onset_d <- merge(assign_tab, onset_d, by = "stay_id")
onset_d$class <- factor(onset_d$class, levels = 1:K)
onset_d$sepsis3 <- as.integer(as.logical(onset_d$sepsis3))
glm_fit <- glm(sepsis3 ~ class, data = onset_d, family = binomial)
glm_s <- summary(glm_fit)$coefficients
onset_tab <- data.frame(term = rownames(glm_s),
                      odds_ratio = round(exp(glm_s[, "Estimate"]), 3),
                      p_value = signif(glm_s[, "Pr(>|z|)"], 3))
cat("\nLogistic model: trajectory class vs incident Sepsis-3 (ref = mildest class):\n")
```

```
##
## Logistic model: trajectory class vs incident Sepsis-3 (ref = mildest class):
```

```

print(onset_tab, row.names = FALSE)

##           term odds_ratio  p_value
## (Intercept)    0.077 8.48e-78
##      class2     6.862 9.84e-37
##      class3    18.809 1.46e-74

write.csv(onset_tab, file.path(OUT, "18_tier6_onset_assoc.csv"), row.names = FALSE)

cat("\nTier 6 (GBTM) complete.\n")

##
## Tier 6 (GBTM) complete.

```